

Data-Based Dynamic Decoupling Control for MIMO Precision Motion Stages With Position-Dependent Disturbances

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Abstract—Decoupling control is a widely employed technique used to mitigate coupling effects and bridge the gap between multiple-input multiple-output (MIMO) and single-input single-output (SISO) control. In the field of precision motion control, the cross-talk resulting from coarse decoupling poses a significant challenge to achieving high performance. Static decoupling control fails to completely decouple the MIMO system due to dynamic disparities between drives, actuators, and the flexible modes of the plant. Consequently, dynamic decoupling control methods have gained attention for their potential to enhance performance. However, existing dynamic decoupling control methods suffer from limitations such as reliance on system models, and susceptibility to disturbances and noise. In this paper, these deficiencies are addressed by 1) a data-based optimization with no involvement of model knowledge, and 2) using the augmented vector and instrumental variable to eliminate the estimation bias caused by position-dependent disturbances and measurement noise. The effectiveness and superiority of the proposed method are substantiated through numerical simulations and experiments conducted on an ultra-precision wafer stage.

Note to Practitioners—This paper presents a non-iterative dynamic decoupling method, which is suitable for LTI multivariable systems with stringent precision requirements, such as the wafer stage and the atomic force microscope. In the field of precision motion control, the feedback controller plays a crucial role in the stability and robustness of the system. However, to achieve the desired performance, additional methods may be necessary,

such as feedforward control method, nonlinear control method, etc. Especially for multiple-input multiple-output systems, it is essential to maintain high decoupling accuracy to avoid the cross-talk, which emerges due to a coarse decoupling. Given the practical differences in dynamic characteristics between drives and actuators, as well as non-rigid modes in the system, dynamic decoupling is recommended over static decoupling. Furthermore, it is promising to develop a data-based method that enables inevitable uncertainties between the model and the actual plant. In the dynamic decoupling method proposed in this paper, the impact of position-dependent disturbances and the measurement noise are jointly taken into consideration, which has not been previously addressed in literature. The augmented vector and instrumental variable are introduced to eliminate the adverse effects of these irrelevant data. Consequently, an unbiased estimate of the dynamic decoupling controller can be obtained. The practicality and effectiveness of the proposed method have been demonstrated. In future work, we will focus on the optimal selection of the dynamic decoupling controller under bounded disturbances and its application to redundant systems.

Index Terms—Cross-talk, data-based, dynamic decoupling control, multiple-input multiple-output (MIMO), position-dependent disturbance.

I. INTRODUCTION

DECOUPLING control is extensively employed in various practical multiple-input multiple-output (MIMO) linear time-invariant (LTI) systems. In these systems, each output is affected by multiple inputs to varying degrees, and each input can exert effects on multiple outputs [1], [2]. To obtain the desired output, a closed-loop control strategy should be utilized, especially for unstable systems where a MIMO feedback controller is required to stabilize the system. However, it is rather complicated to directly design a MIMO centralized controller. For simplicity, the decoupling control method can be used, wherein a decoupling controller is applied to make the decoupled system as diagonal as possible. Subsequently, a decentralized controller can be designed based on the elements of the main diagonal.

In systems with low response speed and control accuracy requirements, such as those observed in chemical processes [3], it is sufficient to employ a static decoupling controller [4]. Similarly, in mechatronic systems with sufficient rigidity, the desired performance can also be obtained by the static decoupling method. However, the static decoupling control becomes unsatisfactory in systems that require fast

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response speed and high precision, such as those found in automotive [5] and power domains [6]. This is particularly true for the wafer stage, which serves as one of the core motion elements in lithographic equipment. To meet the precision and throughput requirements in chip manufacturing, the wafer stage must achieve nanometer-level positioning accuracy within milliseconds after an acceleration of tens of meters per second squared [7]. On the other hand, the lightweight design of the motion stage increases the coupling complexity in the mid- to high- frequency range, thereby reducing the performance of static decoupling. Consequently, the errors caused by other reference inputs, referred to as cross-talk, have become the pivotal factor deteriorating system performance.

Although the cross-talk can be mitigated by increasing the bandwidth of the decentralized controller, the achievable bandwidth is limited by the mechanical structure [8]. Therefore, the most direct and effective approach to addressing this issue is to improve decoupling accuracy.

A significant amount of literature has been presented to obtain an optimal decoupling controller. Previous work has explored the conditions under which a dynamic decoupling controller can achieve full decoupling [9], highlighting that an LTI plant can be decoupled with internal stability under the unity-feedback configuration if there are no coincidences of poles and zeros in the right half-plane [10]. A general form of the decoupling controller derived from the inverse model of the plant has been proposed in [11]. Unfortunately, obtaining an accurate inverse model of a MIMO plant is a challenging task. Attempts to design the decoupling controller using approximations of the inverse model have been made, including the optimal static decoupling method. The elements of static decoupling controller can be optimized through step response experiments [12] with actual steady-state gains between the inputs and outputs, or by using a reference model method.

However, as mentioned above, only low-frequency cross-talk can be eliminated by the static decoupling method. To exceed this limitation, the dynamic decoupling method has attracted significant attention [13]. Various model-based dynamic decoupling control algorithms, including ideal, simplified, and inverted decoupling methods, have been summarized in [14], which still necessitate the use of the inverse model of the plant. Since the modeling error is inevitable, especially for complicated mechatronic systems, the achievable performance of these model-based decoupling methods is limited, which motivates the development of data-based methods.

Compared to model-based methods, data-based methods are more appealing due to their reliance solely on measurement data. Some iterative optimization algorithms have been proposed to update the dynamic decoupling controller parameters by minimizing the norm of the cross-talk. Specifically, a quadratic function of the servo errors of all degrees-of-freedom (DOFs) when controlling a single DOF can serve as an objective function. Accordingly, gradient algorithms can be utilized to minimize the problem [15]. In [16], a modified objective function was introduced to concentrate more on cross-talk elimination. To expedite convergence, an improved gradient method was proposed in [17]. Despite

their usefulness, the measurement data used in these methods are inevitably contaminated by measurement noise, leading to biased decoupling controller parameters and hindering the achievement of the ideal decoupling controller. Several attempts have been made to eliminate the impact of measurement noise, such as the development of iterative feedback tuning (IFT) method [18] where several experiments are performed in each iteration under the assumption that the noise between different trials is uncorrelated. Nevertheless, the measured errors are not exclusively made up of cross-talk and noise, but also errors generated by external disturbances, which most literature ignores. While external disturbances can be significantly compensated using disturbance observer (DOB) [19], iterative learning control (ILC) [20], some residual effects remain. Therefore, it is crucial to mitigate the impact of disturbances when utilizing the measurement data to refine the dynamic decoupling controller.

This paper presents a novel direct data-based decoupling method, inspired by the virtual reference feedback tuning (VRFT) method [21], to address position-dependent external disturbances frequently experienced in real-world systems. Compared to existing literature, the key contributions of this work can be succinctly outlined as follows:

- 1) A data-based dynamic decoupling control method is proposed to improve the decoupling level in high dynamic motion. The dynamics of motors and drives are taken into consideration as well as the flexibility of the mechanical structure. To optimize the parameters, a data-based optimization method is developed, where only a single experiment is required and no need for accurate model information.
- 2) In the presence of position-dependent disturbances and measurement noise, the optimal dynamic decoupling controller can still be determined by introducing the instrumental variable and the augmented vector in the proposed approach, which is a feat not found in previous techniques.
- 3) An application to an ultra-precision wafer stage prototype is provided to highlight the effectiveness and the enhanced performance of the proposed approach over traditional approaches.

II. PROBLEM FORMULATION

A. Description of the Control System

The structure of the wafer stage comprises two components: a long-stroke stage (LS) and a short-stroke stage (SS). The LS offers a broad motion range and ensures micrometer-level precision, while the SS provides a limited motion range with nanometer-level precision. The LS and SS are driven by linear motors and voice coil motors, while the displacements of six DOFs in a Cartesian coordinate fixed to the metrological frame can be measured. In the actual chip manufacturing process, the wafer stage can operate in two modes: stepping motion or scanning motion, and the most critical exposure takes place during the scanning motion. In this scenario, the wafer stage is controlled to move along with the scanning direction, usually coincides with an axis of the measurement coordinate

TABLE I
NOMENCLATURE

Indices		Numeric indices	
t	Period index	i, j	
Sets		Numeric indices	
$\mathbb{R}$	Real number	$\mathbb{R}^n$	Real vector
$\mathbb{C}$	Complex number	$\mathbb{C}^n$	Complex vector
Vector variables		Numeric indices	
$\mathbf{r}(t)$	Reference input	$\mathbf{u}(t)$	Feedback controller output
$\mathbf{f}^u(t)$	Decoupling controller output	$\mathbf{f}^d(t)$	External disturbances
$\mathbf{y}^m(t)$	Measurement output	$\epsilon(t)$	White noise
$\tilde{\mathbf{u}}(t)$	Virtual control input	J, J_i	Criterion function
τ_i^A	$-f_{0,i}^d$	τ_i^B	$-\gamma_{i1}, \gamma_{i2}, \dots, \gamma_{in}]^T$
$\zeta(t)$	Augmented signal	φ_i	Augmented parameter
$k_{ij}, k_{ij}(q^{-1})$	Element of decoupling controller	τ_i	Parameter of i -th column of $\mathbf{K}$
Matrix variables		Numeric indices	
$\mathbf{K}, \mathbf{K}(q^{-1})$	Decoupling controller	$\mathbf{P}, \mathbf{P}(q^{-1})$	Plant model
$\tilde{\mathbf{P}}, \tilde{\mathbf{P}}(q^{-1})$	Diagonal reference model	$\mathbf{C}, \mathbf{C}(q^{-1})$	Feedback controller
Φ	Toeplitz matrix of $\tilde{\mathbf{u}}(t)$	Γ	Parameter matrix of $\mathbf{f}^d$
		$\mathbf{H}, \mathbf{H}(q^{-1})$	Colored noise model
		$\mathbf{F}_i$	Toeplitz matrix of $\mathbf{f}_i(t)$
		Ω	Toeplitz matrix of $\zeta(t)$

system. Meanwhile, other DOFs are controlled to compensate for topography errors, so as to achieve the best exposure result. It is evident that the multiple DOFs motions and high motion performance are required, such that a well-designed decoupling controller should be utilized.

Consider the control configuration of the SS as depicted in Fig. 1. The actual plant, denoted as $\mathbf{P}(q^{-1}) \in \mathbb{C}^{n \times n}$, is assumed to be linear time-invariant and its accurate model is unknown. The symbol q^{-1} denotes the backward shift operator, i.e. $q^{-1}\mathbf{x}(t) = \mathbf{x}(t-1)$. $\mathbf{K}(q^{-1}) \in \mathbb{C}^{n \times n}$ represents the decoupling controller which is used to diagonalize the plant and $\mathbf{C}(q^{-1}) \in \mathbb{C}^{n \times n}$ is a decentralized controller used to stabilize the decoupled system.

The actual output, measurement output and reference input are denoted as $\mathbf{y}(t) \in \mathbb{R}^n$, $\mathbf{y}^m(t) \in \mathbb{R}^n$ and $\mathbf{r}(t) \in \mathbb{R}^n$ respectively. $\mathbf{u}(t) \in \mathbb{R}^n$ represents the control input calculated by $\mathbf{C}(q^{-1})$. The measurement noise $\mathbf{v}(t) \in \mathbb{R}^n$ is assumed to be given by $\mathbf{v}(t) = \mathbf{H}(q^{-1})\epsilon(t)$. The colored noise model $\mathbf{H}(q^{-1}) \in \mathbb{C}^{n \times n}$ is stable and diagonal. $\epsilon(t) \in \mathbb{R}^n$ is normally disturbed white noise vector with zero mean and variance λ_{ϵ}^2 . It is uncorrelated with $\mathbf{r}(t)$, and the elements of $\epsilon(t)$ are mutually uncorrelated. $\mathbf{f}^u(t) \in \mathbb{R}^n$ and $\mathbf{f}(t) \in \mathbb{R}^n$ denote the output of the decoupling controller and the total external force applied to the plant respectively. The input disturbances of the plant, which comprise cable force, the ripple force of the actuator, and son on, is usually position-dependent and can be expressed as $\mathbf{f}^d(t)$. From Fig. 1, the system can be described as

$$\begin{cases} \mathbf{y}(t) = \mathbf{P}(q^{-1})\mathbf{f}(t) \\ \mathbf{f}(t) = \mathbf{f}^d(t) + \mathbf{f}^u(t) \\ \mathbf{f}^u(t) = \mathbf{K}(q^{-1})\mathbf{u}(t) \\ \mathbf{u}(t) = \mathbf{C}(q^{-1})(\mathbf{r}(t) - \mathbf{y}^m(t)) \\ \mathbf{y}^m(t) = \mathbf{y}(t) + \mathbf{v}(t). \end{cases} \quad (1)$$

Consequently, the closed-loop transfer function of the system can be denoted as

$$\mathbf{y}(t) = \mathbf{G}_r(q^{-1})\mathbf{r}(t) + \mathbf{G}_v(q^{-1})\mathbf{v}(t) + \mathbf{G}_d(q^{-1})\mathbf{f}^d(t) \quad (2)$$

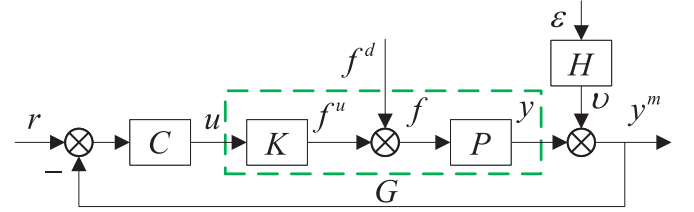


Fig. 1. Block diagram of the typical decoupling control configuration.

where $\mathbf{G}_r(q^{-1}) = [\mathbf{I} + \mathbf{P}(q^{-1})\mathbf{K}(q^{-1})\mathbf{C}(q^{-1})]^{-1}\mathbf{P}(q^{-1})\mathbf{K}(q^{-1})\mathbf{C}(q^{-1})$, $\mathbf{G}_v(q^{-1}) = [\mathbf{I} + \mathbf{P}(q^{-1})\mathbf{K}(q^{-1})\mathbf{C}(q^{-1})]^{-1}$ and $\mathbf{G}_d(q^{-1}) = [\mathbf{I} + \mathbf{P}(q^{-1})\mathbf{K}(q^{-1})\mathbf{C}(q^{-1})]^{-1}\mathbf{P}(q^{-1})$.

Let $\mathbf{S}(q^{-1}) = [\mathbf{I} + \mathbf{P}(q^{-1})\mathbf{K}(q^{-1})\mathbf{C}(q^{-1})]^{-1}$, Eq. (2) can be written as follows in the absence of disturbances and noise

$$\mathbf{y}(t) = \mathbf{G}_r(q^{-1})\mathbf{r}(t) = (\mathbf{I} - \mathbf{S}(q^{-1}))\mathbf{r}(t). \quad (3)$$

In a fully decoupled system, the output of each DOF $y_i(t)$ is uniquely determined by the input of the corresponding DOF $r_i(t)$ with no interference from other DOFs, $i = 1, 2, \dots, n$. This desirable feature can be achieved if and only if $\mathbf{S}(q^{-1})$ is diagonal, as indicated by (3). Alternatively, according to the definition of $\mathbf{S}(q^{-1})$, the ideal decoupling of the system can be obtained when $\mathbf{P}(q^{-1})\mathbf{K}(q^{-1})$ is diagonal. Under the rigid body hypothesis, the plant can be fully decoupled by a static decoupling controller which is a constant matrix. However, due to the inevitable dynamic differences between motors, drives and the flexible modes of the mechanical structure, the effective decoupling can only be realized in low frequency range by the static decoupling method. The movement of any DOF is consequently influenced by the others, leading to the so-called cross-talk. Within the context of wafer stage systems, cross-talk imposes a significant constraint on the system's ability to meet performance requirements.

B. Dynamic Decoupling Control Strategy

To mitigate cross-talk in the mid- to high-frequency range, the static decoupling controller should be replaced with a

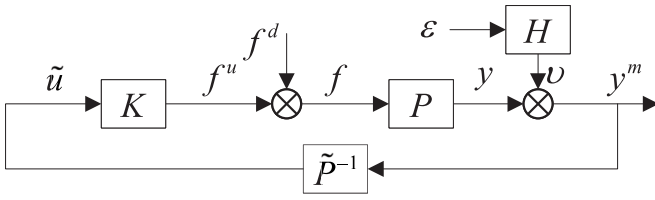


Fig. 2. Block diagram of the proposed approach.

dynamic decoupling controller $\mathbf{K}(q^{-1}) \in \mathbb{C}^{n \times n}$ as follows

$$\mathbf{K}(q^{-1}) = \begin{bmatrix} k_{11}(q^{-1}) & k_{12}(q^{-1}) & \cdots & k_{1n}(q^{-1}) \\ k_{21}(q^{-1}) & k_{22}(q^{-1}) & \cdots & k_{2n}(q^{-1}) \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1}(q^{-1}) & k_{n2}(q^{-1}) & \cdots & k_{nn}(q^{-1}) \end{bmatrix}. \quad (4)$$

Due to the causal and stable system can be accurately approximated by finite-order models structured with finite impulse response (FIR) filter, $k_{ij}(q^{-1})$ in (4) can be represented by an FIR filter with the order of m :

$$k_{ij}(q^{-1}) = \theta_{ij,0} + \theta_{ij,1}q^{-1} + \cdots + \theta_{ij,m}q^{-m} = \boldsymbol{\psi}\boldsymbol{\theta}_{ij} \quad (5)$$

with parameters $\boldsymbol{\theta}_{ij} = [\theta_{ij,0}, \theta_{ij,1}, \dots, \theta_{ij,m}]^T \in \mathbb{R}^{m+1}$ and $\boldsymbol{\psi} = [1, q^{-1}, \dots, q^{-m}]$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$.

The primary objective, therefore, is to optimize the parameters $\boldsymbol{\theta}_{ij}$ in the decoupling controller using measurement data, especially in the presence of measurement noise and position-dependent disturbances. The optimal estimate of $\boldsymbol{\theta}_{ij}$ can be represented as $\boldsymbol{\theta}_{ij}^*$ and the optimal decoupling controller is denoted as $\mathbf{K}^*(q^{-1})$.

III. DATA-BASED DECOUPLING CONTROLLER

A. Proposed Approach in the Absence of Disturbances and Noise

Motivated by VRFT method, the proposed dynamic decoupling control scheme is shown in Fig. 2. First of all, a nominal static decoupling controller $\mathbf{K}_0$ and a decentralized controller $\mathbf{C}(q^{-1})$ are used to stabilize the system. Thus, the variables $\mathbf{r}(t)$, $\mathbf{y}^m(t)$ and $\mathbf{f}^u(t)$ are all available in the absence of disturbances and noise. In order to optimize the decoupling controller $\mathbf{K}(q^{-1})$, a desired diagonal model $\tilde{\mathbf{P}}(q^{-1}) = \text{diag}\{\tilde{p}_1(q^{-1}), \dots, \tilde{p}_n(q^{-1})\} \in \mathbb{C}^{n \times n}$ is given as the reference model. Without loss of generality, the symbol q^{-1} can be omitted, e.g. $\mathbf{P}(q^{-1}) = \mathbf{P}$. Furthermore, the following assumption should hold to apply the method.

Assumption 1: There exists a specific sequence $\tilde{\mathbf{u}}(t)$ which is also referred to as the virtual control input, and when this sequence is applied to the reference model $\tilde{\mathbf{P}}$ as control input, the output of the reference model is equal to $\mathbf{y}(t)$. $\tilde{\mathbf{u}}(t)$ can be determined as

$$\tilde{\mathbf{u}}(t) = \tilde{\mathbf{P}}^{-1}\mathbf{y}(t). \quad (6)$$

It should be emphasized that not every $\tilde{\mathbf{P}}$ is realizable. In fact, the $\tilde{\mathbf{P}}$ is realizable only when certain conditions are satisfied. Specifically, the following conditions must be met:

- 1) $\tilde{\mathbf{P}}$ has no poles in the right half of the s-plane.
- 2) $\tilde{\mathbf{P}}$ is minimum phase.
- 3) The corresponding decoupling controller $\mathbf{K}^*$ is proper [10].

Condition (1) and (2) are easily attainable, and are naturally satisfied in most mechatronic systems with reasonable structure, including the wafer stage examined in this paper. To meet the third condition, the relative order of $\tilde{\mathbf{P}}$ should be greater than or equal to that of $\mathbf{P}$. When these conditions are satisfied, the desired decoupling controller that can decouple the system entirely can be determined using measurement data solely, and the following theorem can be derived.

Theorem 1: Under Assumption 1, the decoupled system is fully diagonal if $\mathbf{f}(t) = \mathbf{K}\tilde{\mathbf{u}}(t)$ is satisfied by the dynamic decoupling controller $\mathbf{K}$.

Proof: From (6), $\mathbf{f}(t) = \mathbf{K}\tilde{\mathbf{u}}(t)$ can be written as

$$\mathbf{f}(t) = \mathbf{K}\tilde{\mathbf{P}}^{-1}\mathbf{y}(t). \quad (7)$$

In the architecture depicted in Fig. 1, $\mathbf{y}(t) = \mathbf{P}\mathbf{f}(t)$. As a result, the following equation can be established

$$\mathbf{y}(t) = \mathbf{P}\mathbf{K}\tilde{\mathbf{P}}^{-1}\mathbf{y}(t), \quad (8)$$

which means $\mathbf{P}\mathbf{K} = \tilde{\mathbf{P}}$.

Therefore $\mathbf{K} = \mathbf{K}^*$ and the closed-loop system can be written as $\mathbf{y}(t) = (\mathbf{I} - \mathbf{S})\mathbf{r}(t) = (\mathbf{I} - (\mathbf{I} + \mathbf{P}\mathbf{K}\mathbf{C})^{-1})\mathbf{r}(t) = (\mathbf{I} - (\mathbf{I} + \tilde{\mathbf{P}}\mathbf{C})^{-1})\mathbf{r}(t)$. Because $\tilde{\mathbf{P}}$ and $\mathbf{C}$ are both diagonal, $(\mathbf{I} - \mathbf{S})$ is diagonal, and thus the decoupled system is fully diagonal. $\square$

Based on Theorem 1, the parameters of $\mathbf{K}$ can be estimated by minimizing a criterion function, defined as follows:

$$J = \sum_{t=1}^N \|\mathbf{f}(t) - \mathbf{K}\tilde{\mathbf{u}}(t)\|_2^2 \quad (9)$$

where $\|\cdot\|_2$ represents the l_2 -norm. $\mathbf{f}(t) = [f_1(t), f_2(t), \dots, f_n(t)]^T \in \mathbb{R}^n$, $\tilde{\mathbf{u}}(t) = [\tilde{u}_1(t), \tilde{u}_2(t), \dots, \tilde{u}_n(t)]^T \in \mathbb{R}^n$.

Substituting (5) into (9) leads to

$$\begin{aligned} J &= \sum_{t=1}^N \|\mathbf{f}(t) - \mathbf{K}\tilde{\mathbf{u}}(t)\|_2^2 \\ &= \sum_{t=1}^N \sum_{i=1}^n (f_i(t) - \sum_{j=1}^n \tilde{u}_j^T(t)\boldsymbol{\theta}_{ij})^2 \end{aligned} \quad (10)$$

where $\tilde{u}_j(t) = \boldsymbol{\psi}^T \tilde{\mathbf{u}}_j(t) = [\tilde{u}_j(t), \tilde{u}_j(t-1), \dots, \tilde{u}_j(t-m)]^T \in \mathbb{R}^{m+1}$.

For simplicity, the following notations are defined.

$$\begin{cases} \mathbf{F}_i = [f_i(1), f_i(2), \dots, f_i(N)]^T \in \mathbb{R}^N \\ \boldsymbol{\tau}_i = [\boldsymbol{\theta}_{i1}^T, \boldsymbol{\theta}_{i2}^T, \dots, \boldsymbol{\theta}_{in}^T]^T \in \mathbb{R}^{(m+1)n} \\ \boldsymbol{\phi}(t) = [\tilde{u}_1^T(t), \tilde{u}_2^T(t), \dots, \tilde{u}_n^T(t)]^T \in \mathbb{R}^{(m+1)n} \\ \Phi = [\boldsymbol{\phi}(1), \boldsymbol{\phi}(2), \dots, \boldsymbol{\phi}(N)]^T \in \mathbb{R}^{N \times (m+1)n} \end{cases} \quad (11)$$

Exchange the order of addition operations, (10) can be written as

$$J = \sum_{i=1}^n \sum_{t=1}^N (f_i(t) - \sum_{j=1}^n \tilde{u}_j^T(t)\boldsymbol{\theta}_{ij})^2$$

$$\begin{aligned}
&= \sum_{i=1}^n (\mathbf{F}_i - \Phi \boldsymbol{\tau}_i)^T (\mathbf{F}_i - \Phi \boldsymbol{\tau}_i) \\
&= \sum_{i=1}^n \|\mathbf{F}_i - \Phi \boldsymbol{\tau}_i\|_2^2.
\end{aligned} \quad (12)$$

Let $J_i = \|\mathbf{F}_i - \Phi \boldsymbol{\tau}_i\|_2^2$, and it is evident that each row of $\mathbf{K}$ can be optimized independently. $\boldsymbol{\tau}_i$ can be estimate by

$$\hat{\boldsymbol{\tau}}_i = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{F}_i \quad (13)$$

where $\hat{\boldsymbol{\tau}}_i = [(\hat{\boldsymbol{\theta}}_{i1})^T, (\hat{\boldsymbol{\theta}}_{i2})^T, \dots, (\hat{\boldsymbol{\theta}}_{in})^T]^T$ and the parameter of k_{ij} can be estimated as $\hat{\boldsymbol{\theta}}_{ij}$. In the absence of disturbances and noise, the optimal parameters of $\mathbf{K}^*$ can be obtained in term of (13), which means

$$\boldsymbol{\tau}_i^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{F}_i \quad (14)$$

where $\boldsymbol{\tau}_i^* = [(\boldsymbol{\theta}_{i1}^*)^T, (\boldsymbol{\theta}_{i2}^*)^T, \dots, (\boldsymbol{\theta}_{in}^*)^T]^T$ for $i = 1, 2, \dots, n$.

However, the external disturbances are unavoidable in actual systems which are difficult to measure and even vary during operation. A large number of existing methods become invalid in the presence of disturbances, and thus, improvements are necessary to mitigate the negative effects on the approach developed in this section.

B. Eliminating The Impact of Disturbances Using Augmented Vector

1) Assessment of the Position-Dependent Disturbances:

This section investigates the properties and effects of external disturbances in the wafer stage system. During operation, external disturbances such as residual gravity, cable force, and stochastic disturbances may arise. Although these can be compensated algorithmically or structurally, some residual disturbance forces remain. Notably, the disturbances are position dependent and can be assumed to exert in the input of the plant. Therefore, the overall input of the plant $\mathbf{f}(t)$ can be expressed as $\mathbf{f}(t) = \mathbf{f}^u(t) + \mathbf{f}^d(t)$, where only the decoupling controller output $\mathbf{f}^u(t)$ is measurable and the disturbance $\mathbf{f}^d(t)$ can not be obtained directly. Subsequently, (13) should be rewritten as

$$\hat{\boldsymbol{\tau}}_i = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{F}_i^u \quad (15)$$

where $\mathbf{F}_i^u = [f_i^u(1), f_i^u(2), \dots, f_i^u(N)]^T \in \mathbb{R}^N$. Let $\mathbf{F}_i^d = [f_i^d(1), f_i^d(2), \dots, f_i^d(N)]^T \in \mathbb{R}^N$, $\mathbf{F}_i^u = \mathbf{F}_i - \mathbf{F}_i^d$ can be derived and (15) can be written as

$$\begin{aligned}
\hat{\boldsymbol{\tau}}_i &= (\Phi^T \Phi)^{-1} \Phi^T (\mathbf{F}_i - \mathbf{F}_i^d) \\
&= \boldsymbol{\tau}_i^* - (\Phi^T \Phi)^{-1} \Phi^T \mathbf{F}_i^d.
\end{aligned} \quad (16)$$

It is evident from (16) that the estimate is biased, highlighting the necessity to eliminate the impact of position-dependent disturbances. To address this issue, an augmented vector is introduced in the following section.

2) *The Augmented Vector Method:* Basically, the residual position-dependent disturbance $\mathbf{f}^d(t)$ can be approximated using Taylor's expansion with respect to $\mathbf{y}(t)$ as follows

$$\mathbf{f}^d(t) = \mathbf{f}_0^d + \Gamma \mathbf{y}(t) + o(1) \quad (17)$$

where $\mathbf{f}_0^d = [f_{0,1}^d, f_{0,2}^d, \dots, f_{0,n}^d] \in \mathbb{R}^n$ and $\Gamma = \{\gamma_{ij}\} \in \mathbb{R}^{n \times n}$, $i, j = 1, 2, \dots, n$ are constant vector and constant matrix respectively. $o(n)$ represents an infinitely small quantity of order n .

To eliminate the impact of disturbances, (9) should be modified based on (17) as follows

$$\begin{aligned}
J &= \sum_{t=1}^N \|\mathbf{f}(t) - \mathbf{K} \tilde{\mathbf{u}}(t)\|^2 \\
&= \sum_{t=1}^N \|\mathbf{f}^u(t) + \mathbf{f}_0^d + \Gamma \mathbf{y}(t) - \mathbf{K} \tilde{\mathbf{u}}(t)\|^2.
\end{aligned} \quad (18)$$

Define $\mathbf{g}(t) = [l(t), l(t), \dots, l(t)] \in \mathbb{R}^n$ where $l(t) = 1$ for $t > 0$ and $l(t) = 0$ for $t \leq 0$. Subsequently, (18) can be written as

$$\begin{aligned}
J &= \sum_{t=1}^N \sum_{i=1}^n (f_i^u(t) + f_{0i}^d l(t) + \sum_{j=1}^n \gamma_{ij} y_j(t) - \boldsymbol{\phi}^T(t) \boldsymbol{\tau}_i)^2 \\
&= \sum_{t=1}^N \sum_{i=1}^n (f_i^u(t) - l(t) \boldsymbol{\tau}_i^A - \mathbf{y}^T(t) \boldsymbol{\tau}_i^B - \boldsymbol{\phi}^T(t) \boldsymbol{\tau}_i)^2
\end{aligned} \quad (19)$$

where $\boldsymbol{\tau}_i^A = -\mathbf{f}_{0,i}^d$ and $\boldsymbol{\tau}_i^B = -[\gamma_{i1}, \gamma_{i2}, \dots, \gamma_{in}]^T \in \mathbb{R}^n$.

Let $\mathbf{p}(t) = [l(t), \mathbf{y}^T(t)]^T \in \mathbb{R}^{n+1}$ which can be referred to as the augmented vector, the impact of the disturbances can be mitigated by using that. For ease of derivation, the following notations are introduced.

$$\begin{cases} \boldsymbol{\zeta}(t) = [\mathbf{p}^T(t), \boldsymbol{\phi}^T(t)]^T \in \mathbb{R}^{(m+1)n+n+1} \\ \boldsymbol{\varphi}_i = [(\boldsymbol{\tau}_i^A)^T, (\boldsymbol{\tau}_i^B)^T, (\boldsymbol{\tau}_i)^T]^T \in \mathbb{R}^{(m+1)n+n+1} \\ \boldsymbol{\Omega} = [\boldsymbol{\zeta}(1), \boldsymbol{\zeta}(2), \dots, \boldsymbol{\zeta}(N)]^T \in \mathbb{R}^{N \times [(m+1)n+n+1]} \end{cases} \quad (20)$$

Substituting (20) into (19), the objective function can be derived as

$$\begin{aligned}
J &= \sum_{i=1}^n \sum_{t=1}^N (f_i^u(t) - \boldsymbol{\zeta}^T(t) \boldsymbol{\varphi}_i)^2 \\
&= \sum_{i=1}^n (\mathbf{F}_i^u - \boldsymbol{\Omega} \boldsymbol{\varphi}_i)^T (\mathbf{F}_i - \boldsymbol{\Omega} \boldsymbol{\varphi}_i) \\
&= \sum_{i=1}^n \|\mathbf{F}_i^u - \boldsymbol{\Omega} \boldsymbol{\varphi}_i\|_2^2.
\end{aligned} \quad (21)$$

Let $J_i = \|\mathbf{F}_i^u - \boldsymbol{\Omega} \boldsymbol{\varphi}_i\|_2^2$, then the compound parameter $\boldsymbol{\varphi}_i$ can be estimated as

$$\hat{\boldsymbol{\varphi}}_i = (\boldsymbol{\Omega}^T \boldsymbol{\Omega})^{-1} \boldsymbol{\Omega}^T \mathbf{F}_i^u. \quad (22)$$

The parameters $\boldsymbol{\tau}_i$ of decoupling controller $\mathbf{K}$ can be obtained from $\hat{\boldsymbol{\varphi}}_i = [(\hat{\boldsymbol{\tau}}_i^A)^T, (\hat{\boldsymbol{\tau}}_i^B)^T, (\hat{\boldsymbol{\tau}}_i)^T]^T$. Compared with (13), the identification model in (22) includes an augmented vector $\mathbf{p}(t)$ to eliminate the effect of disturbances. In the absence of noise, accurate parameter estimation of the decoupling controller can be obtained using (22), which can be represented as

$$\boldsymbol{\tau}_i^* = (\boldsymbol{\Omega}^T \boldsymbol{\Omega})^{-1} \boldsymbol{\Omega}^T \mathbf{F}_i^u.$$

$$\begin{aligned}
&= [(\tau_i^A)^*; (\tau_i^B)^*; (\tau_i)^*] \\
&= [-(f_{0i}^d)^*, -\gamma_{i1}^*, \dots, -\gamma_{in}^*, (\theta_{i1}^*)^T, \dots, (\theta_{in}^*)^T]^T \quad (23)
\end{aligned}$$

for $i = 1, 2, \dots, n$.

However, measurement noise is inevitably and the consistency of (23) must be validated.

C. Eliminating Impact of Noise Using Instrumental Variable

1) *Consistency Analysis*: In this section, the consistency analysis of the proposed method is carried out. The measurement output $\mathbf{y}^m(t)$ is contaminated by noise inevitably, therefore it can be written as

$$\begin{aligned}
\mathbf{y}^m(t) &= \mathbf{G}_r^y \mathbf{r}(t) + \mathbf{G}_l^y f_0^d l(t) + \mathbf{G}_v^y \mathbf{v}(t) \\
&= \mathbf{y}^0(t) + \mathbf{y}^v(t) \quad (24)
\end{aligned}$$

where $\mathbf{G}_r^y = (\mathbf{I} + \mathbf{PKC} - \mathbf{P}\Gamma)^{-1}\mathbf{PKC}$, $\mathbf{G}_l^y = (\mathbf{I} + \mathbf{PKC} - \mathbf{P}\Gamma)^{-1}\mathbf{P}$ and $\mathbf{G}_v^y = (\mathbf{I} + \mathbf{PKC} - \mathbf{P}\Gamma)^{-1}(\mathbf{I} - \mathbf{P}\Gamma)$. $\mathbf{y}^v(t) = \mathbf{G}_v^y \mathbf{v}(t)$ is denoted as the output correlated with $\mathbf{v}(t)$, and $\mathbf{y}^0(t) = \mathbf{G}_r^y \mathbf{r}(t) + \mathbf{G}_l^y f_0^d l(t)$ represents the output of the system without noise.

Since the measurement output is $\mathbf{y}^m(t)$, the actual $\tilde{\mathbf{u}}(t)$ is determined as

$$\begin{aligned}
\tilde{\mathbf{u}}(t) &= \tilde{\mathbf{P}}^{-1} \mathbf{y}^m(t) \\
&= \tilde{\mathbf{P}}^{-1} \mathbf{y}^0(t) + \tilde{\mathbf{P}}^{-1} \mathbf{y}^v(t) \\
&= \tilde{\mathbf{u}}^0(t) + \tilde{\mathbf{u}}^v(t). \quad (25)
\end{aligned}$$

Analogously, the output of decoupling controller $\mathbf{f}^u(t)$ is affected by the noise $\mathbf{v}(t)$ due to the closed loop, which can be described as

$$\begin{aligned}
\mathbf{f}^u(t) &= \mathbf{G}_r^f \mathbf{r}(t) + \mathbf{G}_d^f (f_0^d l(t) + \Gamma(\mathbf{y}^m(t) - \mathbf{v}(t))) \\
&\quad + \mathbf{G}_v^f \mathbf{v}(t) \quad (26)
\end{aligned}$$

where $\mathbf{G}_r^f = (\mathbf{I} + \mathbf{KCP})^{-1}\mathbf{KC}$, $\mathbf{G}_d^f = (\mathbf{I} + \mathbf{KCP})^{-1}\mathbf{KCP}$ and $\mathbf{G}_v^f = -(\mathbf{I} + \mathbf{KCP})^{-1}\mathbf{KC}$. From (24) to (26), $\mathbf{f}^u(t)$ is comprised of two components $\mathbf{f}^0(t) = \mathbf{G}_r^f \mathbf{r}(t) + \mathbf{G}_d^f (f_0^d l(t) + \Gamma \mathbf{y}^0(t))$ and $\mathbf{f}^v(t) = \mathbf{G}_v^f \mathbf{v}(t) + \mathbf{G}_d^f (\Gamma \mathbf{y}^v(t) - \mathbf{v}(t))$, while $\mathbf{f}^v(t)$ is correlated with noise.

Let $\tilde{\mathbf{u}}_j^0(t) = \boldsymbol{\psi}^T \tilde{\mathbf{u}}_j(t) = [\tilde{u}_j^0(t), \tilde{u}_j^0(t-1), \dots, \tilde{u}_j^0(t-m)]^T \in \mathbb{R}^{m+1}$ and $\tilde{\mathbf{u}}_j^v(t) = \boldsymbol{\psi}^T \tilde{\mathbf{u}}_j^v(t) = [\tilde{u}_j^v(t), \tilde{u}_j^v(t-1), \dots, \tilde{u}_j^v(t-m)]^T \in \mathbb{R}^{m+1}$. Hence, $\mathbf{y}^m(t)$, $\mathbf{f}^u(t)$ and $\tilde{\mathbf{u}}(t)$ can be described by

$$\begin{cases} \mathbf{y}^m(t) = \mathbf{y}^0(t) + \mathbf{y}^v(t) \\ \mathbf{f}^u(t) = \mathbf{f}^0(t) + \mathbf{f}^v(t) \\ \tilde{\mathbf{u}}(t) = \tilde{\mathbf{u}}^0(t) + \tilde{\mathbf{u}}^v(t) \end{cases} \quad (27)$$

Substituting (27) into the definition of $\boldsymbol{\xi}(t)$, we have

$$\begin{aligned}
\boldsymbol{\xi}(t) &= [l(t), y_1^m(t), \dots, y_n^m(t), \tilde{\mathbf{u}}_1^T(t), \dots, \tilde{\mathbf{u}}_n^T(t)]^T \\
&= \boldsymbol{\xi}^0(t) + \boldsymbol{\xi}^v(t) \quad (28)
\end{aligned}$$

where $\boldsymbol{\xi}^0(t) = [l(t), y_1^0(t), \dots, y_n^0(t), (\tilde{\mathbf{u}}_1^0(t))^T, \dots, (\tilde{\mathbf{u}}_n^0(t))^T]^T \in \mathbb{R}^{(m+1)n+n+1}$ and $\boldsymbol{\xi}^v(t) = [0, y_1^v(t), \dots, y_n^v(t), (\tilde{\mathbf{u}}_1^v(t))^T, \dots, (\tilde{\mathbf{u}}_n^v(t))^T]^T \in \mathbb{R}^{(m+1)n+n+1}$.

Consequently, Ω and $\mathbf{F}_i$ can be represented as

$$\begin{cases} \Omega = \Omega^0 + \Omega^v \\ \mathbf{F}_i = \mathbf{F}_i^0 + \mathbf{F}_i^v \end{cases} \quad (29)$$

where

$$\begin{cases} \Omega^0 = [\boldsymbol{\xi}^0(1), \boldsymbol{\xi}^0(2), \dots, \boldsymbol{\xi}^0(N)]^T \in \mathbb{R}^{N \times [(m+1)n+n+1]} \\ \Omega^v = [\boldsymbol{\xi}^v(1), \boldsymbol{\xi}^v(2), \dots, \boldsymbol{\xi}^v(N)]^T \in \mathbb{R}^{N \times [(m+1)n+n+1]} \\ \mathbf{F}_i^0 = [f_i^0(1), f_i^0(2), \dots, f_i^0(N)]^T \in \mathbb{R}^N \\ \mathbf{F}_i^v = [f_i^v(1), f_i^v(2), \dots, f_i^v(N)]^T \in \mathbb{R}^N \end{cases} \quad (30)$$

Subsequently, the estimate of (23) can be written as

$$\begin{aligned}
\hat{\boldsymbol{\phi}}_i &= (\Omega^T \Omega)^{-1} \Omega^T \mathbf{F}_i^u \\
&= ((\Omega^0 + \Omega^v)^T (\Omega^0 + \Omega^v))^{-1} (\Omega^0 + \Omega^v)^T (\mathbf{F}_i^0 + \mathbf{F}_i^v) \\
&= ((\Omega^0)^T \Omega^0 + (\Omega^v)^T \Omega^v)^{-1} ((\Omega^0)^T \mathbf{F}_i^0 + (\Omega^v)^T \mathbf{F}_i^v). \quad (31)
\end{aligned}$$

As the noise $\mathbf{v}(t)$ is colored noise, $(\Omega^v)^T \Omega^v \neq \mathbf{0}$ and $(\Omega^v)^T \mathbf{F}_i^v \neq \mathbf{0}$, which leads to biased estimate. To obtain consistent estimate result, an improved method is proposed in the following section.

2) The Instrumental Variable Method for Noise Rejection:

This section presents an improved approach that utilizes instrumental variable to eliminate the influence of measurement noise. According to (23) and (27), the following equation is satisfied when the optimal estimate of $\boldsymbol{\phi}_i^*$ can be obtained for $i = 1, 2, \dots, n$.

$$\mathbf{f}^0(t) + (\mathbf{f}_0^d)^* l(t) + \Gamma^* \mathbf{y}^0(t) - \mathbf{K}^* \tilde{\mathbf{u}}^0(t) = \mathbf{0} \quad (32)$$

where $(\mathbf{f}_0^d)^* = [(f_{0,1}^d)^*, \dots, (f_{0,n}^d)^*]^T$, $\Gamma^* = \{\gamma_{ij}^*\}$ and $\mathbf{K}^* = \{\boldsymbol{\psi} \theta_{ij}^*\}$. Define the augmented error as follows

$$\rho(t) = \mathbf{f}^u(t) + \mathbf{f}_0^d l(t) + \Gamma \mathbf{y}^m(t) - \mathbf{K} \tilde{\mathbf{u}}(t) \quad (33)$$

The main idea is to adjust the parameters of $\mathbf{f}_0^d$, Γ and $\mathbf{K}$ to ensure that the augmented error $\rho(t)$ is uncorrelated with the instrumental variable $\mathbf{z}(t) \in \mathbb{R}^{nm+2n+1}$, which is designed to be related to the reference input and unrelated to the measurement noise. Consequently, the following theorem can be derived:

Theorem 2: Consider the MIMO system shown in Fig. 1, and given the measurement data $\mathbf{f}^u(t)$ and $\mathbf{y}^m(t)$ for $t = 1, 2, \dots, N$. Then

$$\lim_{N \rightarrow \infty} \hat{\boldsymbol{\phi}}_i = \left(\frac{1}{N} \sum_{t=1}^N \mathbf{z}(t) \boldsymbol{\xi}^T(t) \right)^{-1} \frac{1}{N} \sum_{t=1}^N \mathbf{z}(t) \mathbf{f}_i^u(t) = \boldsymbol{\phi}_i^* \quad (34)$$

for $i = 1, 2, \dots, n$ if $E[\mathbf{z}(t) \boldsymbol{\xi}^T(t)]$ is nonsingular and $E[\mathbf{z}(t) \mathbf{v}_j(t)] = 0$ for $j = 1, 2, \dots, n$, where $\mathbf{z}(t) \in \mathbb{R}^{(m+1)n+n+1}$ is deterministic.

Proof: Substituting (28) and (27) into (34), the estimate becomes

$$\begin{aligned}
\hat{\boldsymbol{\phi}}_i &= \left[\frac{1}{N} \sum_{t=1}^N \mathbf{z}(t) ((\boldsymbol{\xi}^0(t))^T + (\boldsymbol{\xi}^v(t))^T) \right]^{-1} \\
&\quad \cdot \frac{1}{N} \sum_{t=1}^N \mathbf{z}(t) (f_i^0(t) + f_i^v(t)). \quad (35)
\end{aligned}$$

Since the instrumental variable $\mathbf{z}(t)$ is uncorrelated with the noise, it can be concluded that

$$\begin{cases} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbf{z}(t) (\boldsymbol{\xi}^v(t))^T = \mathbf{0} \\ \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbf{z}(t) f_i^v(t) = 0 \end{cases} \quad (36)$$

Therefore, Eq. (35) can be written as

$$\hat{\phi}_j = \left[\frac{1}{N} \sum_{t=1}^N z(t)(\zeta^0(t))^T \right]^{-1} \frac{1}{N} \sum_{t=1}^N z(t)f_i^0(t). \quad (37)$$

From (23), in absence of noise, $f_i^0(t) - (\zeta^0(t))^T \phi_i^* = 0$ can be derived. Substituting it into (37), the estimate can be written as

$$\begin{aligned} \hat{\phi}_i &= \left[\frac{1}{N} \sum_{t=1}^N z(t)(\zeta^0(t))^T \right]^{-1} \frac{1}{N} \sum_{t=1}^N z(t)(\zeta^0(t))^T \phi_i^* \\ &= \phi_i^*. \end{aligned} \quad (38)$$

□

Consequently, the unbiased estimate of θ_{ij}^* can be obtained using (34).

IV. SIMULATIONS AND EXPERIMENTS

To demonstrate the effectiveness of the proposed method, particularly its capability to mitigate the impact of position-dependent disturbances and measurement noise, three decoupling control methods are compared, which are outlined as follows:

- 1) M_1 : Static decoupling control method proposed in [22].
- 2) M_2 : Preexisting data-based iterative dynamic decoupling method proposed in [18].
- 3) M_3 : Proposed data-based method with instrumental variable.

The decoupling controllers obtained using different methods are denoted as $\mathbf{K}_1$, $\mathbf{K}_2$ and $\mathbf{K}_3$ respectively.

A. Simulation Results

In this section, a simulation example of a two-input two-output (TITO) system is presented. Suppose that the inputs corresponding to the current commands of the drives, while the outputs are denoted as the displacements of the two DOFs, the accurate plant model can be represented as

$$\begin{aligned} \mathbf{P}(s) &= \mathbf{P}_0(s)\mathbf{D}_m(s) \\ &= \begin{bmatrix} \frac{0.08}{s^2} & \frac{0.5333}{s^2} \\ \frac{0.1067}{s^2} & \frac{-0.4}{s^2} \end{bmatrix} \cdot \begin{bmatrix} \frac{0.000606s+1}{0.000202s+1} & 0 \\ 0 & \frac{0.000603s+1}{0.000201s+1} \end{bmatrix}. \end{aligned} \quad (39)$$

where $\mathbf{P}_0(s)$ and $\mathbf{D}_m(s)$ represent the rigid body model and the dynamics of motors respectively.

According to (39), it is evident that achieving fully decoupling of $\mathbf{P}(s)$ via a static decoupling controller is impossible. In traditional decoupling method, the dynamics differences are disregarded such that a static decoupling controller can be determined. Given the desired diagonal model $\mathbf{P}^*$, the ideal static decoupling controller can be determined as

$$\mathbf{K}_s = (\mathbf{P}_0(s))^{-1} \cdot \mathbf{P}^*(s)|_{s=0}. \quad (40)$$

However, due to uncertainties, the precise parameters of the plant are unknown, and only a nominal model of the plant is available before stabilizing the system. Therefore, only an

TABLE II
MEANINGS AND PROPOSED VALUES OF THE PID CONTROLLER

Parameter	Meaning	Proposed value
K_p	Proportional gain	$m(2\pi f_{bw})/\alpha$
f_0	Integrator frequency	f_{bw}/α^2
f_1	Differentiator frequency	f_{bw}/α
f_2	Low-pass filter frequency	$f_{bw}\alpha$
ξ	Low-pass filter damping ratio	0.707
m	Equivalent mass to be moved	10
f_{bw}	Desired bandwidth (Hz)	30
α	PID frequency ratio	3

nominal static decoupling controller can be derived based on the nominal model as follows

$$\begin{aligned} \mathbf{K}_1 &= \hat{\mathbf{P}}_0(s)^{-1} \cdot \mathbf{P}^*(s)|_{s=0} \\ &= \begin{bmatrix} \frac{0.1}{s^2} & \frac{0.5}{s^2} \\ \frac{0.1}{s^2} & \frac{-0.5}{s^2} \end{bmatrix}^{-1} \cdot \begin{bmatrix} \frac{0.1}{s^2} & 0 \\ 0 & \frac{0.1}{s^2} \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & -0.1 \end{bmatrix} \end{aligned} \quad (41)$$

where $\hat{\mathbf{P}}_0(s)$ is the nominal model which is not equal to the actual model $\mathbf{P}_0(s)$.

By using the nominal decoupling controller $\mathbf{K}_1$, the plant can become diagonal dominant. Subsequently, a decentralized control method can be utilized to stabilize the system. The feedback controller can be designed in term of the diagonal elements of the desired decoupled system, which can be expressed as

$$\mathbf{C}(s) = \begin{bmatrix} C_1(s) & 0 \\ 0 & C_2(s) \end{bmatrix}. \quad (42)$$

where $C_1(s)$ and $C_2(s)$ are PID controllers. The criteria introduced in [22] can be used to determine the controller parameters. In brief, the PID controller to stabilize a plant with double integral properties can be parameterized as

$$\begin{aligned} C_i(s) &= K_p \cdot \frac{s + 2\pi f_0}{s} \cdot \left(\frac{s}{2\pi f_1} + 1 \right) \\ &\quad \cdot \frac{(2\pi f_2)^2}{s^2 + 2\xi(2\pi f_2)s + (2\pi f_2)^2}. \end{aligned} \quad (43)$$

for $i = 1, 2$. The meanings and the proposed values of the parameters can refer to Table II.

In this example, the equivalent masses of the desired decoupled system are both equal to ten. The bandwidth f_{bw} is chosen as 30 Hz. It should be note that although the the decentralized controller is designed based on the desired decoupled model, which may differ from the actual system, the closed system can still maintain stability.

Besides, the discrete form of $\mathbf{P}(s)$ and $\mathbf{C}(s)$ are used with the sample time of $T_s=200 \mu s$, and denoted as $\mathbf{P}(q^{-1})$ and $\mathbf{C}(q^{-1})$ respectively. The position-dependent disturbances are assumed to be of linear form which can be represented as

$$\mathbf{f}^d(t) = \begin{bmatrix} f_1^d(t) \\ f_2^d(t) \end{bmatrix} = \begin{bmatrix} f_{10}^d + f_{11}^d y_1(t) + f_{12}^d y_2(t) \\ f_{20}^d + f_{21}^d y_1(t) + f_{22}^d y_2(t) \end{bmatrix}. \quad (44)$$

TABLE III
PARAMETERS OF POSITION DEPENDENT DISTURBANCES

Parameter	Value (m)	Parameter	Value (m)
f_{10}^d	0.2	f_{20}^d	-0.3
f_{11}^d	1000	f_{21}^d	300
f_{12}^d	500	f_{22}^d	-1000

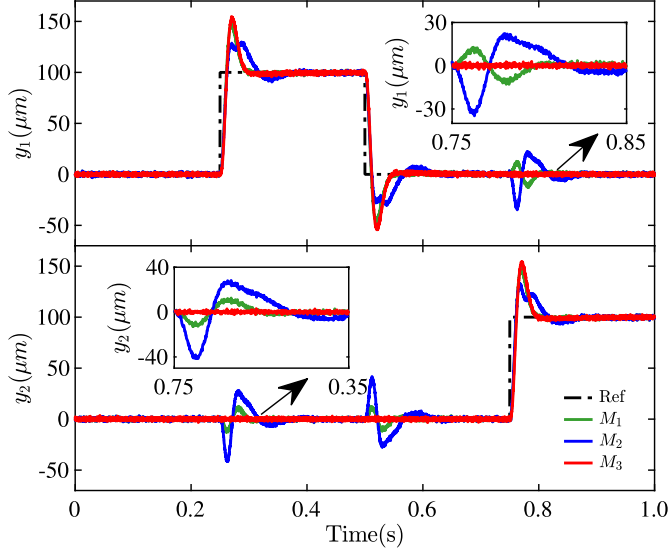


Fig. 3. Close-loop responses with different decoupling controllers. The black dotted line represents the reference trajectory. The green line, blue line and red lines represent the output responses of the systems determined by M_1 , M_2 and M_3 respectively.

where $y_1(t)$ and $y_2(t)$ are outputs of the system. The values of these parameters used in simulations are listed in Tab. III.

The measurement noise $\mathbf{v}(t) = \mathbf{H}(q^{-1})\epsilon(t)$ is colored noise where $\mathbf{H}(q^{-1}) = \text{diag}\{h(q^{-1}), h(q^{-1})\}$ and $\epsilon_i(t)$, $i = 1, 2$, is Gaussian white noise with the variance of 10^{-12} . In this simulation, the noise model is designated as

$$h(q^{-1}) = \frac{1 - 1.571q^{-1} + 0.6481q^{-2}}{0.8048 - 1.61q^{-1} + 0.8048q^{-2}} \quad (45)$$

and $\epsilon_1(t)$ is uncorrelated to $\epsilon_2(t)$.

In M_2 , the order of the FIR filter is set to six, and convergence is attained after 15 iterations. In M_3 , the decoupling controller is defined by (4) with the order of k_{ij} set to six, i.e., $m = 6$. The excitation signals $r_1(t)$ and $r_2(t)$ are selected as stochastic M-sequences, which are uncorrelated to each other. Consequently, two dynamic decoupling controllers $\mathbf{K}_2$ and $\mathbf{K}_3$ can be obtained through the M_2 method and the M_3 method respectively.

To demonstrate the effectiveness of the proposed method, tracking experiments are conducted where two square waves are exerted to the systems with different decoupling controllers in absence of disturbances, which implies $\mathbf{f}^d(t) = \mathbf{0}$. In the tracking experiments, all outputs are controlled to 0 μm from 0 s to 0.25 s. Then, y_1 is controlled to step 100 μm and held until 0.5 s. Subsequently, all outputs are returned to 0 μm again until 0.75 s, after which y_2 is controlled to step 100 μm and held until 1 s. The output responses, the control signals of

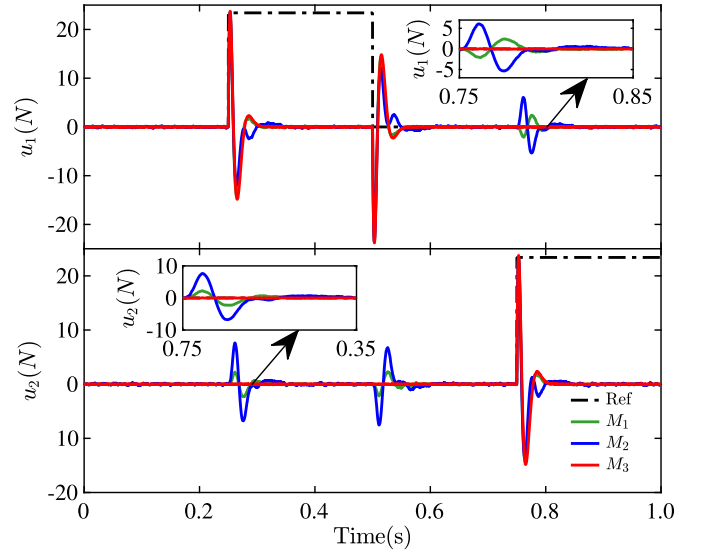


Fig. 4. Control signals of the feedback controllers in systems with different decoupling controllers.

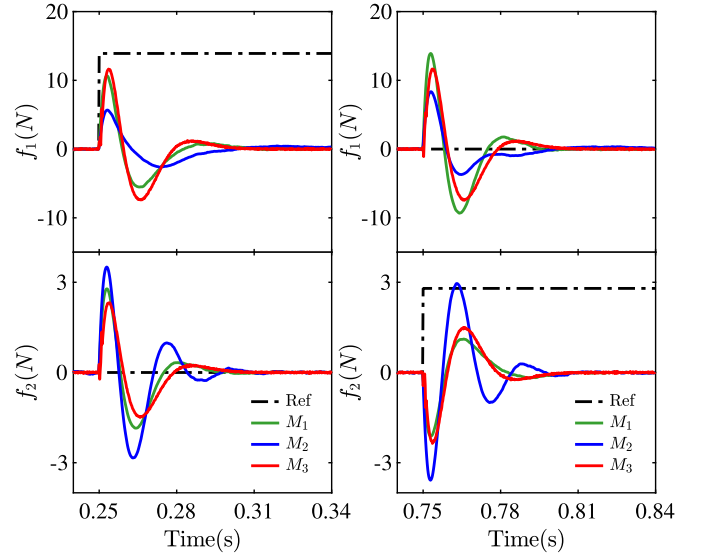


Fig. 5. Control inputs of the plant in tracking experiments with different decoupling controllers.

feedback controllers and the output of decoupling controller (which are indeed the control input of plant) are collected and presented in Fig. 3, Fig. 4 and Fig. 5 respectively. Based on the results, the following conclusions can be drawn:

- 1) From Fig. 3, it is obvious that the nominal decoupled system is not diagonal, and interactions deteriorate the performance significantly. The maximum cross-talk of y_2 reaches to 12.8 μm when y_1 steps. Similarly, as y_2 steps, the maximum cross-talk of y_1 is up to 12.9 μm . Therefore, it is necessary to improve the decoupling level to achieve the desired performance.
- 2) To enhance the decoupling performance, M_2 is used to determine a dynamic decoupling controller. However, due to the impact of position-dependent disturbances, the parameters are biased. The interactions among the system with $\mathbf{K}_2$ are exacerbated, and the maximum

cross-talk that occur in y_1 and y_2 are $34.8 \mu\text{m}$ and $41.5 \mu\text{m}$ respectively. Consequently, the tracking performance of y_1 and y_2 has suffered, as evident in Fig. 4.

- 3) Comparing with M_1 and M_2 , the proposed data-based decoupling method M_3 can achieve the most significant performance improvement. The interactions are almost completely eliminated, which indicate that the decoupled system is nearly diagonal. As a consequence, the effectiveness of the proposed method in improving the decoupling level in the presence of disturbances and measurement noise is clearly demonstrated.

B. Experiment Results

In this section, the proposed method is applied to a three-DOFs precision motion stage developed in our laboratory to further verify its effectiveness.

1) *Experimental Setup*: The experimental setup, depicted in Fig. 6, comprises a multi-DOFs precision motion stage, a set of drives, an encoder counter card, a DA transform card, and a professional high-performance controller. The multi-DOFs precision motion stage consists of a stator, which is fixed to the base, and a mover. These two components are interconnected via three elastic connectors, which serve to constrain the horizontal displacement of the mover. Three voice coil motors are utilized to actuate the mover, with the steel components of the motors installed in the mover and the coils mounted on the stator. Therefore, the mover can achieve three DOFs displacements in vertical direction by controlling three motors to output different driving forces. According to the coordinate defined as Fig. 6 shows, the three DOFs of motion can be denoted as z -DOF, R_x -DOF and R_y -DOF. The z -DOF corresponds to motion along the vertical direction, while the R_x -DOF and R_y -DOF represent rotation around the x and y axes respectively. Suppose the controlled point is the center of the mover, the displacements along the three DOFs in the coordinate system C_{xyz} can be determined based on the measurement values of four linear encoders.

These linear encoders are capable of measuring the vertical movement of their respective installation locations, with a resolution of 5 nm . Through finite element simulation software calculations, the mass of the mover, and the moment of inertia for R_x -DOF and R_y -DOF are $m = 4.5862 \text{ kg}$, $J_{R_x} = 0.0484 \text{ kg m}^2$ and $J_{R_y} = 0.0414 \text{ kg m}^2$ respectively. The drives are selected as commercial DC amplifier amplifiers, model TA115, with a gain of 0.2 A/V . The digital signals from the linear encoders are captured by the encoder counter card. Subsequently, the effective measurement data is transmitted to the controller for the calculation of displacements. The control program is implemented in the high-performance integrated controller where a TMS320C6678 DSP chip is used as main computing unit. To achieve excellent performance, the closed loop frequency is up to 5 kHz , which means the sampling time $T_s = 200 \mu\text{s}$. The discrete controller $\mathbf{C}(\mathbf{q}^{-1})$ is implemented in the controller, and the digital control signal is sent to the DA transform module. Consequently, the voice coil motors generate driving forces proportional to the driving current, achieving the movement of the mover. The gain of the DA

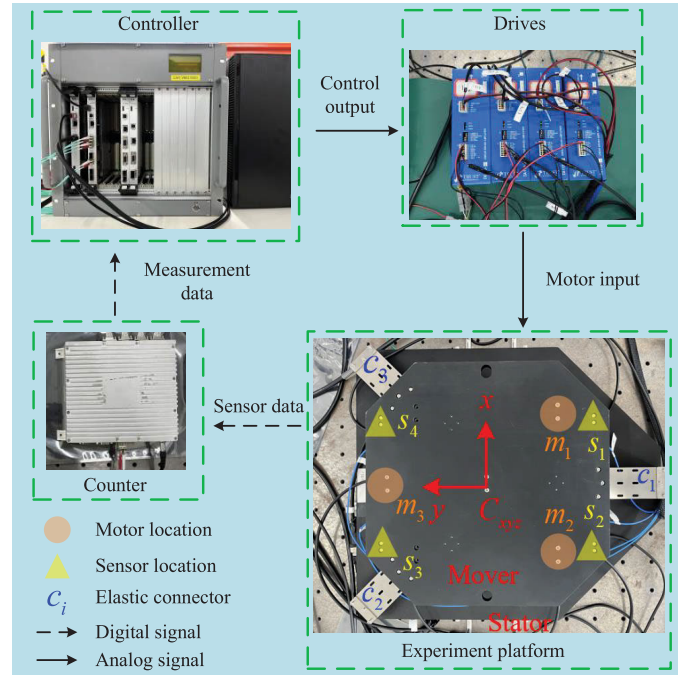


Fig. 6. Experiment setup comprised of a validation prototype experiment platform and corresponding control system hardware.

transform module is 52428.7 V^{-1} and the force constant of the voice coil motor is 12.9 N/A .

Firstly, the nominal static decoupling method is employed to stabilize the system. The desired decoupled model is designed as follows

$$\tilde{\mathbf{P}}(s) = \begin{bmatrix} \frac{1}{4.5862s^2} & 0 & 0 \\ 0 & \frac{1}{0.0484s^2} & 0 \\ 0 & 0 & \frac{1}{0.0414s^2} \end{bmatrix}. \quad (46)$$

Subsequently, a nominal static decoupling controller can be determined which can be represented as

$$\mathbf{K}_1 = \begin{bmatrix} 0.25 & -2.3256 & -4.7619 \\ 0.25 & -2.3256 & 4.7619 \\ 0.5 & 4.6512 & 0 \end{bmatrix}. \quad (47)$$

According to the frequency response information of the plant, the bandwidth of the decentralized controller is chosen as 30 Hz . Following the instruction listed in Tab. II, the feedback controller can be obtained as

$$\mathbf{C}(s) = \begin{bmatrix} C_z(s) & 0 & 0 \\ 0 & C_{R_x}(s) & 0 \\ 0 & 0 & C_{R_y}(s) \end{bmatrix}. \quad (48)$$

where the elements can be detailed as

$$\begin{cases} C_z(s) = \frac{3.53 \times 10^{14}s^2 + 2.957 \times 10^{16}s + 4.645 \times 10^{17}}{62.83s^3 + 5.024 \times 10^4 + 2.009 \times 10^7} \\ C_{R_x}(s) = \frac{3.725 \times 10^{12}s^2 + 3.121 \times 10^{14}s + 4.902 \times 10^{15}}{62.83s^3 + 5.024 \times 10^4 + 2.009 \times 10^7} \\ C_{R_y}(s) = \frac{3.186 \times 10^{12}s^2 + 2.669 \times 10^{14}s + 4.193 \times 10^{15}}{62.83s^3 + 5.024 \times 10^4 + 2.009 \times 10^7} \end{cases} \quad (49)$$

Consequently, the prototype is stabilized and three DOFs can be controlled to track designated trajectories. However,

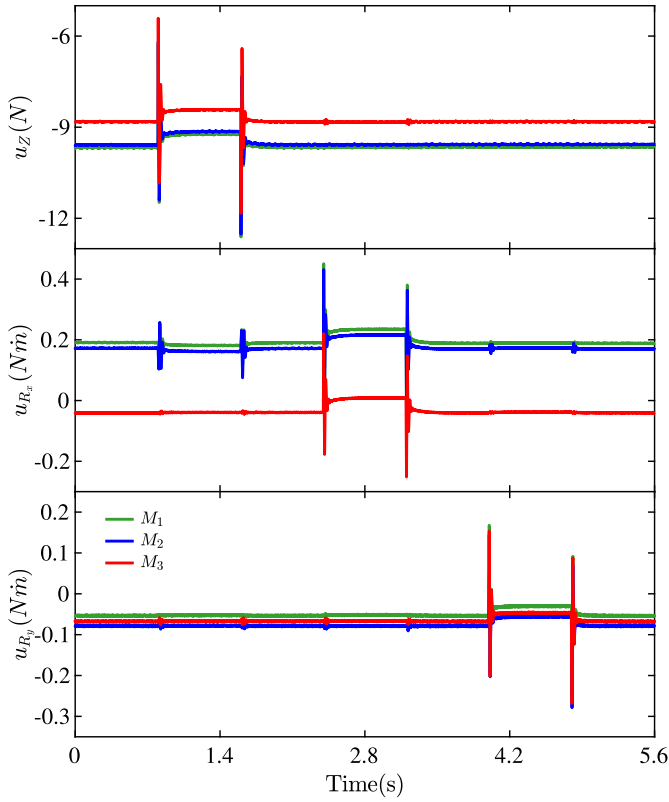


Fig. 7. The control signals of the feedback controllers in tracking experiments.

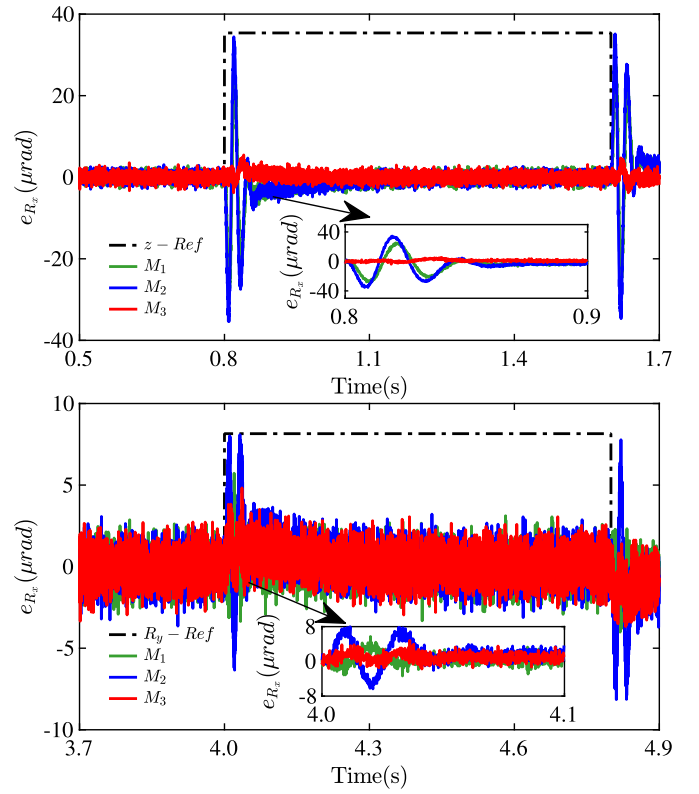


Fig. 9. The cross-talk in R_x -DOF when z -DOF and R_y -DOF step.

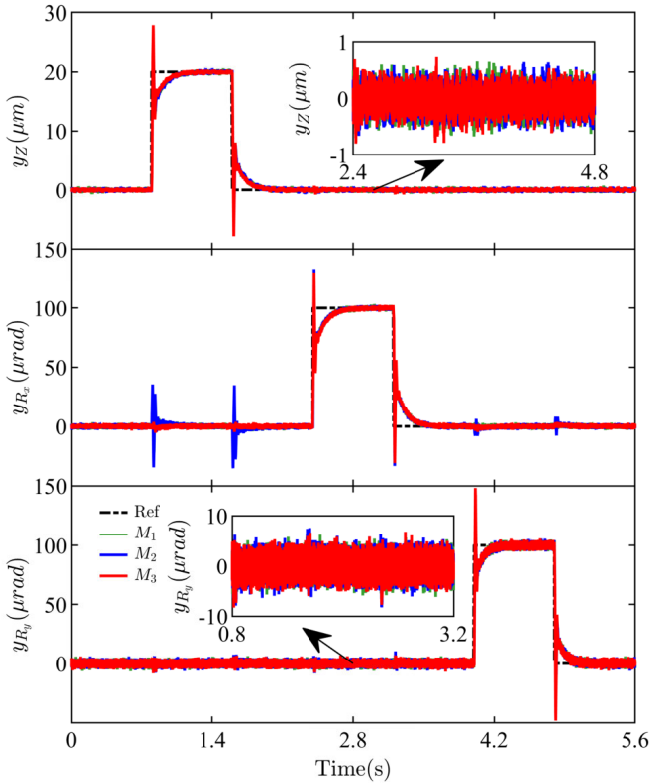


Fig. 8. The output responses of the system in tracking experiments.

as per the previous analysis, due to the manufacturing and assembly errors, the nominal static decoupling controller cannot achieve full diagonal. Furthermore, the differences

between motor dynamics and drives, as well as the distribution of flexible modes, contribute to the deterioration of decoupling and motion performance. To improve the decoupling level, M_2 and the proposed method M_3 are sequentially applied to the system.

In M_2 , the reference signal is selected as a step signal with an amplitude of 20 μm , and it can be observed that the experimental results diverge due to disturbances. In M_3 , the excitation signals are chosen as the M-sequences. The magnitudes of the excitation signals for three DOFs are 5 μm , 20 μrad and 20 μrad respectively. As a result, two decoupling controllers K_2 and K_3 , both with the order of five, are obtained using M_2 and M_3 respectively. To validate the effectiveness of the proposed method, a series of step tracking experiments are carried out. In the tracking experiments, z -DOF, R_x -DOF and R_y -DOF are controlled to step 20 μm , 100 μrad and 100 μrad respectively.

The control signals of the feedback controller and the output responses with different decoupling controllers are presented in Fig. 7 and Fig. 8 respectively. From Fig. 7, it is evident that severe position-dependent disturbances are present in the system, which arise from the elastic connectors between the mover and the stator. When the mover is controlled to different set-points, the feedback controller must generate additional significant control outputs to reject the disturbances. From Fig. 8, it can be observed that the primary interactions are characterized by the influences of the z -DOF and R_y -DOF on R_x -DOF, indicating that couplings between these DOFs can affect system performance. When z -DOF steps or R_y -DOF steps, considerable errors occur in R_x -DOF. Therefore,

TABLE IV

MAXIMUM VALUES OF THE CROSS-TALK IN R_x -DOF WHEN z -DOF AND R_y -DOF STEP WITH DIFFERENT DECOUPLING CONTROLLERS

DOF excited	M_1 (μrad)	M_2 (μrad)	M_3 (μrad)
z	27.858	35.380	5.4639
R_y	5.724	8.088	4.8256

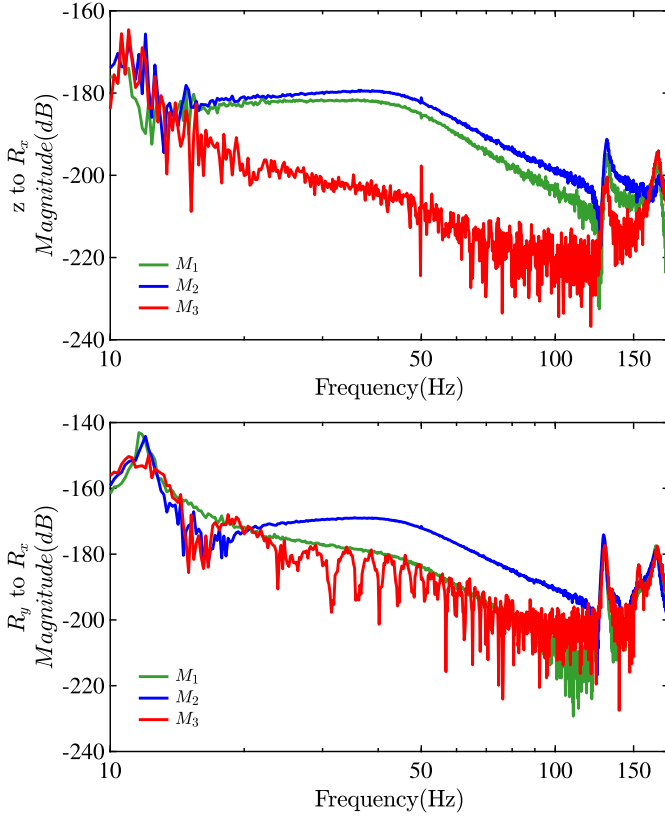


Fig. 10. The frequency response test results from z -DOF to R_x -DOF and from R_y -DOF to R_x -DOF.

to clearly show the effects of different decoupling methods, the cross-talk presented in R_x -DOF can serve as main evaluation criterion, as shown in Fig. 9. The maximum values of the cross-talk occurring in R_x -DOF when z -DOF and R_y -DOF step are listed in Tab. IV.

2) *Effectiveness of the Proposed Method:* Based on the experiment results, the following conclusions can be obtained:

- 1) In comparison with M_1 , a considerable performance improvement is achieved by the proposed method M_3 , as evidenced by the red line in Fig. 9. The cross-talk in R_x -DOF caused by the motion of z -DOF and R_y -DOF is increasingly eliminated. Particularly for the interaction between z -DOF and R_x -DOF, nearly 80% of performance improvement is achieved. These results demonstrate the superiority of the proposed method over traditional ones in effectively decoupling the system and enhancing its performance.
- 2) Compared with M_3 , M_2 is susceptible to unavoidable disturbances, and the decoupling level is further deteriorated by the obtained decoupling controller. This

highlights the necessity of taking disturbances into consideration when optimizing the decoupling controller. Compared to M_2 , the ability of the proposed method to eliminate the impact of disturbances is demonstrated.

- 3) To further illustrate the enhanced performance of the proposed method, the frequency responses of the decoupled systems are tested, and the results are displayed in Fig. 10. The top diagram depicts the frequency responses between the input of z -DOF and the output of R_x -DOF, while the bottom diagram presents the frequency responses between the input of R_y -DOF and the output of z -DOF. According to the definition of the frequency response, it can be concluded that the smaller the magnitude, the weaker the interactions between the two DOFs. This observation further verifies the effectiveness of the proposed method.

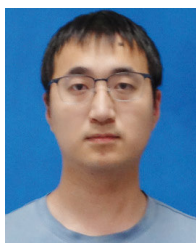
V. CONCLUSION

The decoupling level is essential for achieving the desired performance in high-precision multiple degrees-of-freedom motion systems. Due to the inevitable errors in mechanical parameters and the complex dynamics, the static decoupling method is insufficient to meet the increasingly stringent performance requirement. Accordingly, a dynamic decoupling controller is required, and several methods have been proposed. However, disturbances are neglected in existing literature, even though they are common in practical applications. In this paper, the impact of position dependent disturbances is taken into consideration and an improved data-based decoupling control method is proposed. The effect of disturbances is mitigated by using the augmented vector, and additionally, the instrumental variable is employed to reduce the impact of the measurement noise. Comparative simulation and experimental results confirm the superiority and effectiveness of the proposed method over existing data-based methods. Consequently, this research has broad application prospects and value. Especially in the field of high-precision mechatronic systems, such as atomic force microscopes and lithography. Nevertheless, several problems remain to be addressed, and further work should focus on optimizing the reference model and generalizing it to non-minimum phase systems. It is also challenging work to perform this methodology in multivariable processes with larger differences between the elements of the system model.

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